An Advanced Tree-Algorithm with Free Access, Multipacket Reception and Successive Interference Cancellation

Author 1, Author 2, Author 3 and Author 4

***Abstract*—**

I. Introduction

Massive Internet-of-Things (mIoT) is one of the principal service categories within the scope of the current 5G standardizaton [1]. mIoT is characterized by sporadic uplink transmissions of short data packets by a massive number of devices. This necessitates use of random access procedure each time a device has data to send, either to obtain the resources reserved for sending the data in grant-based access, or to immediately send data in grant-free access. Traditional random access protocols, like ALOHA [2] or tree-algorithms (TA) [3], [4] suffer from modest performance, which in case of massive user arrivals represents a major bottleneck [5].

The performance of random access protocols can be significantly pushed by exploiting advanced physical layer techniques that efficiently deal with collisions among packets of accessing users. One such technique is multipacket reception (MPR), recently shown to significantly increase the maximum stable throughput (MST) of tree-algorithms [6], [7].

Another, related technique is successive interference cancellation (SIC)[1], in which decoding of a user packet enables removal of all other replicas of the same packet that occurred previously[2]. It was demonstrated that SIC also improves the MST of tree-algorithms [8]. An obvious question is whether MPR and SIC can work together to push the MST performance further? In this respect, our focus is on tree algorithms with free access (TAFA), which have been considerably less investigated than the variants with the gated or the windowed access [4], despite their simplicity.

In this letter, we push the state-of-the-art and derive MST for the binary TAFA with MPR and SIC. Specifically, we:

- We propose two new TAFA protocols with MPR and SIC: (i) the first protocol, denoted by Variant A, applies propagation of SIC only backward along the tree, (ii) the second protocol, Variant B, allows for the propagation of SIC both backward and forward along the tree.
- We analyze the MST performance of Variant A using the theory of multitype branching processess (MTP). We show that the obtained MST performance surpasses the best known results known so far TA protocols, even without MPR capability included. When MPR is included, we show that the MST increases further as the MPR capability grows, i.e., as the number of collided packets that can be simultaneously decoded by the receiver increases. Importantly, this also holds for the normalized MST, obtained by assuming the model in which the number of time-frequency resources required to achieve simultaneous decoding of up to and including K packets (i.e., K-MPR) linearly increases with K. Other metrics.
- As the exact analysis of Variant B is intractable, we characterize an upper limit on the performance of Variant B and show that it can not be worse than the one of Variant A. We also compare the performance of two protocols in terms of average delay as function of the arrival intensity, and show that Variant B consistently outperforms Variant A....

The organization of the rest of the text is as follows. We conclude this section with a brief overview of the background and the related work. Section II introduces the system model. Sections III and IV present the analysis. The performance evaluation is presented in Section V. Section VI concludes the letter.

Backgound and related work

The analogies among MTP and TAFA were first identified in [9] and further exploited in [10], [11]. Specifically, the latter work applied the theory of MTP to calculate the MST of various versions of TAFA. Among these, the authors considered a variant of binary TAFA with SIC in which the receiver is able to store the contention outcome of just of the slot that last occurred on the link (denoted as the single-slot memory feature), effectively prohibiting the propagation of the SIC along the tree. The MST of this algorithm was shown to be equal to 0.5699. This is significantly better than the best performing algorithm without SIC - the modified binary TAFA whose MST is 0.4068 [12]. In contrast, the performance of binary TAFA with SIC and single slot memory is notably worse than the binary TA with gated access and SIC, whose MST is 0.6931 [8].

The impact of K-MPR on the performance of binary TAFA was also investigated in [10]. It was shown that MST increases with K, however, when normalized with K, this increase is modest. For instance, when $K = 2$, the normalized MST is 0.3721, and when $K = 10$, the MST is 0.4163.

[1]Strictly speaking, SIC is just another form of MPR. In this paper, we use the term SIC to denote the application of interference cancellation across slots, while we assume that MPR operates on individual slot basis.

[2]Note that retransmissions (i.e., sending of multiple packet replicas) are the inevitable consequences of collisions.

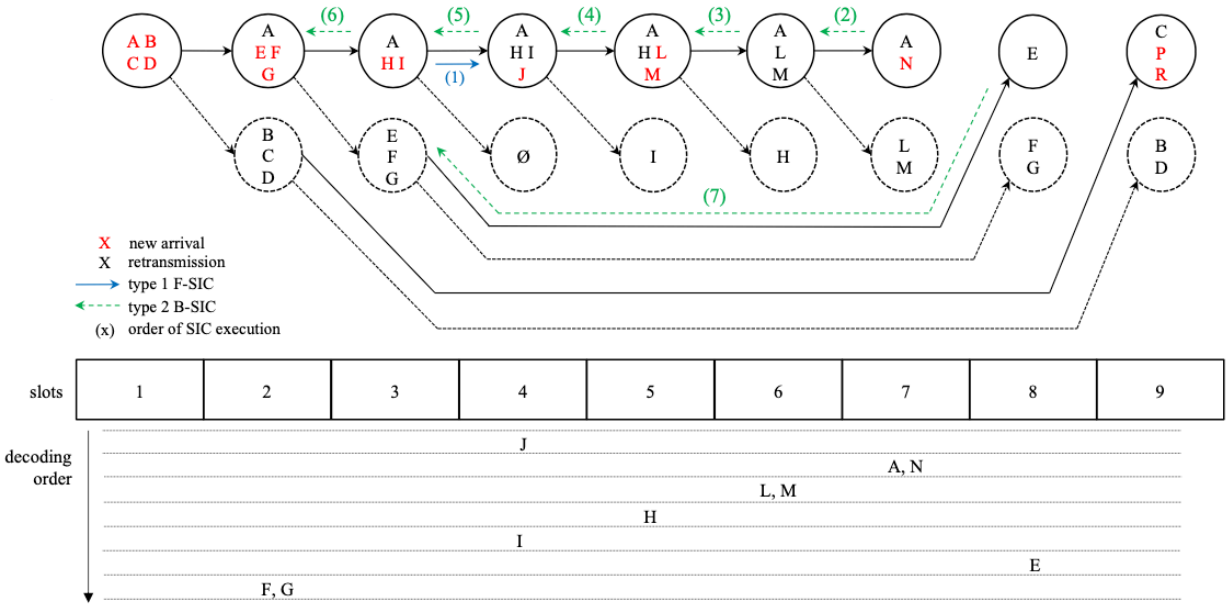


Fig. 1. An example of Variant A protocol with $K = 2$. In slot 1, users A, B, C and D transmit; their collision is not decodable and they split: A chooses the first group and B, C and D the second. A retransmits in slot 2, along with newly arrived users E, F and G. Slot 2 is not decodable and the users that transmitted in it split: A choses the first group, while E, F and G choose the second. A retransmits in slot 3, along with the newly arrived users H, I. Slot 3 is not decodable, and the users transmitting in it split, such that all of them chose the first group and retransmit in slot 4. Via type-1 F-SIC, user J gets decoded in slot 4, while A, H and I remain in the slot slot and split. A and H choose the first group, while I choses the second. A and H retransmit in slot 5 together with the newly arrived L and M; again the slot is not decodable and they split, such that A, L and M choose the first group and H the second. A, L and M retransmit in slot 6, in which there were no new arrivals. Slot 6 is not decodable, the users that transmitted in it split: A chooses the first group, and L and M the second. A retransmits in slot 7 together with the newly arrrived N. This slot is decodable, which triggers a series of type-2 B-SIC and decodings along the tree: A's is removed from slot 6, which becomes decodable; A, L and M are remved from 5, which becomes decodable, A, H get removed from slot 4, which becomes decodable, and A, H and I get removed from slot 3, and finally, A gets removed from slot 2 and slot 1. Now is the turn for the second group {E,F,G}, created after undecodable collision in slot 2 to split. E chooses the newly created first group and F, G and the second. E retransmits in slot 8, and as there were no new arrivals, gets decoded. This decoding event triggers type-2 B-SIC, leading to decoding of F and G. In the next step, users from the second group {B,C,D} created after the undecodable collision in slot 1 split, such that C chooses group 1 and B and D the second. C retransmits in slot 9 together with newly arrived P and R. The protocol operation continues.

In this letter, we consider a general version of the algorithm, i.e., the receiver is able to story any amount of previously occurring collisions until they are resolved.[3] We show that such achieved gains promote the scheme into the best performing TA so far, as well as that the addition of MPR considerably boosts the performance.

Finally, we remark that the computation of the expected delay of successful packet delivery and the expected number of transmission attempts for TAFA via branching processes was presented in [11]. In this paper, we compute these metrics for the scheme considered in this paper leveraging the results presented in [11].

II. System Model

We assume a slotted-time system with an infinite user population. The system is characterized with Poisson arrivals, such that every new packet arrival occurs at a new user and the arrival intensity is λ packets (i.e., users) per slot.[4] The system also features backlogged users, whose transmissions were previously unsuccessful and that are retransmitting according to the protocol elaborated in the further text.

[3]As a consequence, a straightforward generalization of the analytical results presented in [10] is not possible to the problem considered in this paper.

[4]In further text, we will use terms packet decoding and user decoding interchangeably.

The users are slot-synchronized and contend in the uplink for the access to a common access point (AP), which is capable of K-MPR. Each packet transmission also contains: (i) the pointer to the slot in which the packet was sent for the first time, and (ii) the seed of the random number generator used during the splitting process. These pieces of information will enable the AP to figure out in which slots the decoded users (re)transmitted, so that it can perform the SIC.

Every user experiencing a packet arrival sends the packet in the next slot that appears on the channel. In other words, the user is free to transmit as soon as it gets a packet, hence the name of the access policy. These transmissions happen concurrently with the retransmissions from a subset of the backlogged users that were unsuccessful in the previous slot. We denote the original received signal in slot s_i by S_i and the total number of packets contained in S_i, i.e., its *original* degree by $|S_i|$. Via application of SIC, as elaborated later, some of the packets contained in s_i may get removed, such that the degree of the slot becomes reduced.

If $|S_i| > K$, the AP can not immediately decode any packet in slot s_i. All such slots are stored in the AP's memory until all the packets contained in them become decoded, which happens via application of the SIC. Specifically, we consider *forward* SIC (F-SIC) and *backward* SIC (B-SIC), and each of these are further divided into two types. Type-1 B-SIC refers to

the scenario in which the AP performs the operation $S_{i-1} - S_i$. In other words, via type-1 B-SIC, the AP subtracts the complete signal received in s_i from the one received in s_{i-1}, in hope that this will enable decoding of some of the users in s_i, as elaborated later. Type-2 B-SIC refers to the scenario in which the AP performs the operation $S_i - \sum_{k \in \mathcal{D}_j(i)} X_k$, where $D_j(i)$ is the set of user packets decoded in s_j, $i < j$, and whose replicas are also present in s_i, and where X_k are the respective signals. In other words, via type-2 B-SIC the AP removes the decoded packets individually and in an informed way. Analogously, type-1 F-SIC refers to the operation $S_i - S_{i-1}$ and type-2 F-SIC to $S_j - \sum_{k \in \mathcal{D}_i(j)} X_k$, where $i < j$.

We now proceed by elaborating the two algorithms, which differ only in the types of SIC that they exploit. The first algorithm, labeled as Variant A, exploits type-2 B-SIC, and type-1 F-SIC, and the second, labeled as Variant B, exploits all four types of SIC. Obviously, Variant B is the general one and the introduction of Variant A may seem unnecessary. However, the seemingly superficial differences among the two variants allow for the exact analytical treatment of Variant A, which is not the case for Variant B, as elaborated in Section **??**. The protocol operation continues by retransmissions of the users from the last backlogged group 2 as well as the transmissions from the newly arrived users in the next slot.

Variant A

Assume that the AP has just observed the contention outcome in s_i and assume that $|S_i| > K$. If $S_{i-1} > K$ (i.e., the previous slot could also not be decoded), the AP first performs type-1 F-SIC by computing $S_i - S_{i-1}$ and attempts decoding of the obtained signal. This will lead to decoding of newly arrived users in s_i (if there were such) in the case that all the users from s_{i_1} retransmitted in s_i and that $|S_i - S_{i-1}| \leq K$.[5]. An example of such scenario is presented in Fig. 1, user J gets decoded in slot 5, after type-1 F-SIC was applied.

In the next step, the AP stores slot S_i (or the remaining signal if type-1 F-SIC was applied) in the memory and notifies the users from s_i to split into two groups, where each of them randomly and independently joins the first group, denoted by $s_{i,1}$, with probability p, $p \in (0, 1)$, and the second group, denoted by $s_{i,2}$, with probability $1 - p$. The users in the first group retransmit in the next slot, slot s_{i+1}, jointly with any other user that newly arrived during slot s_i. The users in the second group become backlogged and wait for their turn. The backlogging is performed in a stack manner, i.e, the last backlogged group will be treated first when the occasion arises. Also, it is important to note that the users from the second group will *either* become decoded through SIC, *or* they will randomly split again into two new groups (the reason for this will become apparent soon). In the latter case, the users that have chosen the first group will retransmit in the next slot.

Alternatively, if $|S_i| \leq K$, then the AP is able to decode the users that transmitted in s_i. Moreover, some of the decoded users from s_i may have transmitted in some previously memorized slots (the AP learns that via the first-slot pointers and seeds of random number generators contained in the decoded packets), and the AP performs type-2 B-SIC on these slots, removing the transmission of such users. This is illustrated in Fig. 1 in slot 7. Also, since the retransmissions are always performed from the users that ended up in a first group (as indicated before), application of type-2 B-SIC always leaves the users from the second group in the affected slots. These users may get decoded if there are no more than K of them in a (second) group, propelling news round of type-2 B-SIC and potential decodings. This is illustrated in Fig. 1, where the application of type-2 B-SIC allows for decodings of slots 6, 5, and 4. If there are more than K, then these users immediately split, as the AP is already in a possession of their contention outcome. In Fig. 1, this happens in slot 2. Finally, when the successions of SIC and decoding finish, the users from the second group that was created in the last undecoded slot retransmit in the next slot, jointly with the newly arrived users. This is depicted in Fig. 1, where the in users in the second group created in slot 2 split again, and the ones that gave chosen the newly created first group retransmit in slot 8.

Variant B

We describe Variant B by focusing on the differences to the Variant A. Similarly as in Variant A, Variant B starts with type-1 F-SIC among slots s_i and s_{i-1}, by computing $S_i - S_{i-1}$. If the decoding of $S_i - S_{i-1}$ fails, then the AP performs type-1 B-SIC by computing $S_{i-1} - S_i$ and attempts decoding of this signal. This will lead to decoding of some of the users from s_{i-1} if there were no new arrivals in s_i (i.e., s_i contains only retransmissions from a subset of users from s_{i-1}) and if $1 \leq |S_{i-1} - S_i| \leq K$. Some of the decoded users from s_{i-1} may have transmitted in some previous slots and the AP performs type-2 B-SIC on these slots, removing the transmission of such users. Then the AP attempts decoding of the affected slots, which, if successful, will trigger new rounds of *both* type-2 B-SIC (i.e., backwards along the tree) and type-2 F-SIC (i.e., forward along the tree) and potentially lead to new decodings. This is depicted in Fig. 1, in which the type-1 B-SIC enables decoding of user H in slot 5, triggers type-2 B-SIC and decoding of slot 4, which then triggers both type-2 B-SIC and type-2 F-SIC etc. The process ends when no new decodings are possible.

In the next step, if $|S_i| \geq K$ (or if the degree remaining signal in s_i is still greater than K if SIC was applied), the AP stores the signal in the memory and notifies the users from s_i to split into two groups, as in Variant A.

Finally, if $|S_i| \leq K$, the AP decodes the users from s_i, and applies type-2 B-SIC. If some of the decoded users from s_i transmitted in some previously memorized slots, and the AP performs type-2 B-SIC on these slots, removing the transmission of such users. This may lead to new decodings, after which both type-2 B-SIC and type-2 F-SIC are performed, potentially triggering new rounds of decodings, etc. This is illustrated in Fig. 3.

Finally, we remark that the complete protocol should also specify feedback and state variables that will allow the users

[5]This way of exploiting interference cancellation, proposed in [10], is not possible in TA with gated and windowed access, as no new arrivals are allowed in the system during the collision resolution procedure

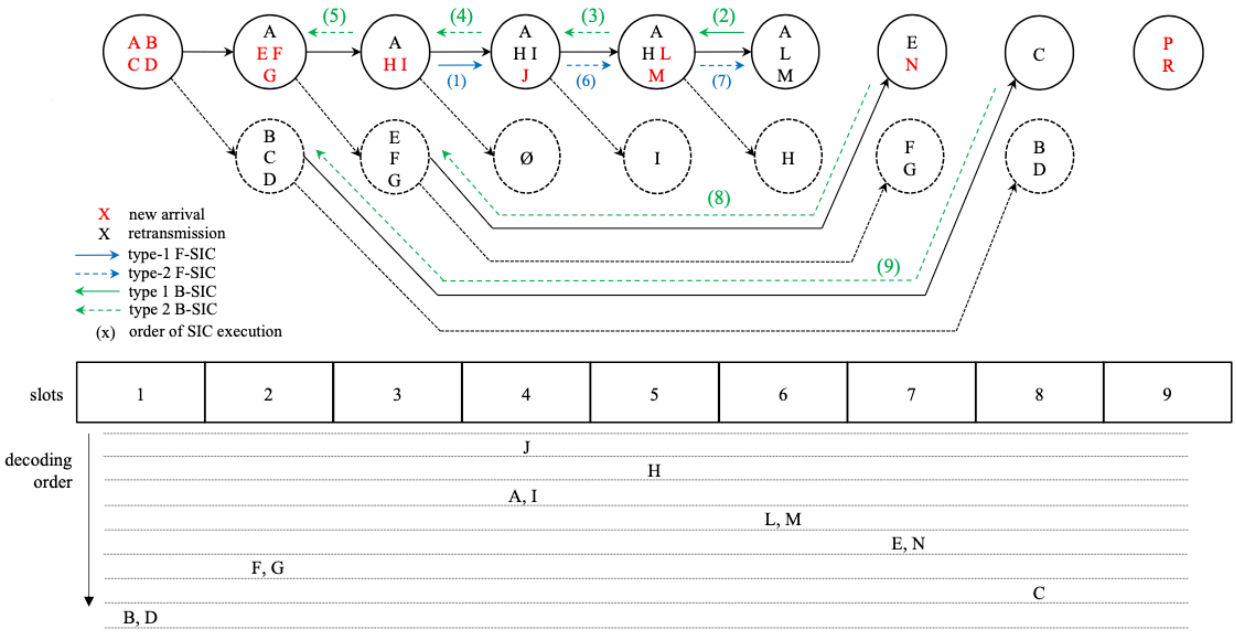


Fig. 2. An example of Variant B protocol with $K = 2$. The example unfolds in the same way as the example in Fig. 1 up to slot 6, when the AP performs type-1 B-SIC and manages to decode user H from slot 5. This triggers type-2 B-SIC, which enables decoding of slot 4 (recall that J was decoded previously after type-1 F-SIC was applied). In turn, this triggers new rounds of both type-2 B-SIC and type-2 F-SIC, enabling decoding of slots 3, 5 and 6. The protocol operation continues by splitting and decoding of the backlogged second groups {E,F,G} and then {B,C,D}.

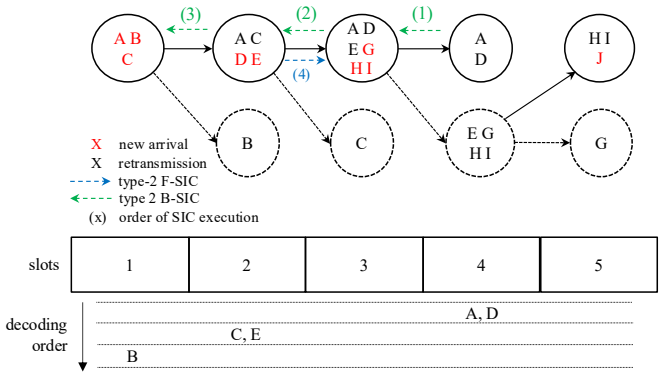


Fig. 3. Another example of Variant B protocol with $K = 2$. The first decoding occurs in slot 4, which triggers type-2 B-SIC along the tree. Eventually type-2 B-SIC reaches slot 1, enabling its decoding and triggering type-2 F-SIC. Slot 2 becomes decoded, which triggers type-2 F-SIC and removal of user E from slot 3 and thus from the second group {E,G,H,I} created in that slot. G, H and I split, such that H and I choose the first group and retransmit in slot 5 together with the newly arrived user J, while G chooses the second group and becomes backlogged. Note that in Variant A, there would be no type-2 F-SIC and, thus, E would remain in the second group {E,G,H,I}.

to learn if and when they should split and contend again, but omit these details due to the space constraints.

III. Analysis of Variant A

In order to analyze the performance of Variant A protocol, we apply the theory of multitype branching processes (MTP). Here we briefly outline their most important notions and refer to the interested reader to [10], [13], for more details. A MTP models evolution of the population of individuals classified in $d + 1$ distinct types over generations, represented by the vector $\mathbf{Z}(n) = [Z_0(n), Z_1(n), \dots Z_d(n)]$ where $Z_i(n)$ is the number of individuals of type i, $0 \leq i \leq d$, in generation n. In each generation, a parent of type i, $0 \leq i, j \leq d$, can produce a number c_j of children of any type j, $0 \leq j \leq d$, with probability $p_{i,c_0,c_1,\dots,c_d}$ that is independent from other parents and their types and that does not change over generations. Moreover, the sequence of vectors $\mathbf{Z}(n)$, $n = 0, 1, 2, \dots$ defines a Markov chain, i.e., exhibits a Markovian property.

For the purposes in this paper, it is sufficient to characterize the MTP through the mean number of children of type j that a parent of type i has in a generation, denoted by $m_{i,j}$. Specifically, if the spectral radius $\rho(\mathbf{M})$ of the matrix $\mathbf{M} = [m_{i,j}]$ is less or equal to 1 ($\mathbf{M}$ should also be non-negative and irreducible), then the MPT is (sub-)critical and will become extinct in a finite amount of generations with probability 1.

In the context of TAFA, individuals of the MTP are collisions, their types correspond to the collision degree, and a generation (i.e., set of children) comprises all the collisions that directly arise from their parent collisions. For instance, in Fig. 1, slot 2 represents an example of a parent collision of degree 4 and its children are collisions in slot 3 (of degree 2) and slot 8 (of degree 1). Another example is the collision of degree 3 in slot 6 that has only one child - the collision of degree 2 in slot 7, as the users that chose the second group in slot 6 were decoded after the application of type-2 B-SIC. The degree of the child collision j depends on the degree of the parent collision i, splitting probability p and the number of newly arrived users (i.e., on the arrival intensity λ). Thus, $\rho(\mathbf{M}) = f(p, \lambda)$ and the goal is to find the combination of p and λ for which $\rho(\mathbf{M}) = 1$. In this case the MST is reached, i.e., the arrival intensity λ becomes the $\lambda_{\max}$.

There are also few important differences between TAFA with SIC and MPR and a standard MTP that we outline here and that moderately modify the application of MTP theory to the problem. Firstly, parent collisions of degree up to and including K will not produce a new generation (i.e., they will have no children), as they will be immediately decoded (i.e., become extinct). Secondly, children collisions of degree up to and including K will also be immediately decoded, thus also not giving rise to a new generation. Finally, degrees of both parent and children collision in our system model are in principle unbounded, as the number of newly arrived users can be arbitrarily high with non-zero probability. However, we set the limit to the maximum collision degree to some large value d for which probability that the collision degree is higher than d is small enough not to impact the analysis [10]. As a consequence, matrix $\mathbf{M}$ in our case contains entries $m_{i,j}$, $K < i, j \leq d$, i.e., it is of order $d - K$.

At this point, we comment why the MTP-based analysis can be applied only to Variant A protocol. Recall that the Markovian property of MTP requires that it suffices to know only the types of the parents in order to determine the types of the children. In Variant A, the original degree of a child collision (i.e., the degree of the child collision when it appears on the channel), depends on the original degree of the parent collision, the outcome of the splitting procedure (which depends on the splitting probability), and on the number of new arrivals which are given by a realization of the stationary arrival process that is memoryless. In effect, this implies that the Markovian property holds. The application of type-1 F-SIC is performed only between the parent and the child collisions and can affect only the degree of the child, thus preserving the Markovian property. Further, a child collision gets immediately decoded (i.e., decoded when it appears on the channel) if and only if its degree is less or equal to K, irrespective of the degree of the parent collision, which is a memoryless feature and as such does not impact the Markovian property. Finally, type-2 B-SIC ensures that the second group will either be immediately decoded (if its size is not more than K) or split (if its size is more than K) which is also a memoryless feature and does not impact the Markovian property.

In Variant B, the application of type-1 B-SIC and/or type-2 F-SIC can cause the changes of the original degrees and/or decoding of the children collisions due to decodings of the collisions that have occurred *before* their parents. An example of the former can be seen in Fig. 3, where the decoding of slot 2 and the consequential application of type-2 F-SIC eliminated user E from the stored slot 4, and thus from the second group created in that slot. This makes the original degree of the collision in slot 5, which is the child of the parent collision in slot 4, to become 3, and in effect demonstrates that the system has a memory of more than a single slot, violating the Markovian property. An example of the latter can be seen in Fig. 1, where the child collision in slot 6 gets decoded, although its original degree is 3, because of the decoding in slot 4, which again demonstrates that the system has a memory of more than a single slot and which violates the Markov property.

We now derive the mean number of children of degree j,

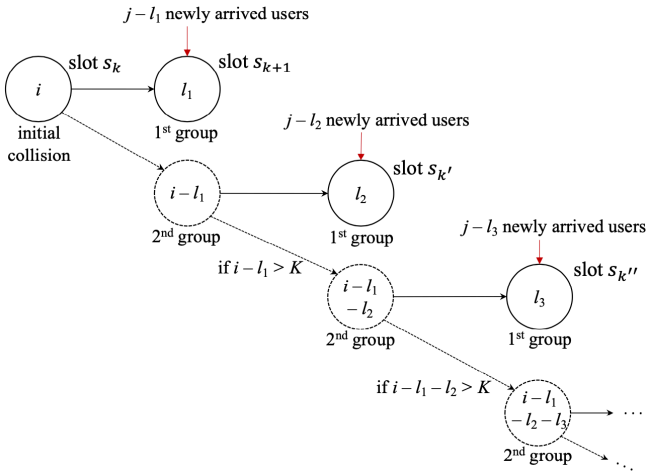


Fig. 4. The potential ways to obtain a child collision of degree j from a parent collision of degree i, $i > K$.

$j > K$, that a parent collision of degree i, $i > K$, in slot s_k, produces in the next generation in Variant A. We first focus on the first child of the parent collision that is created immediately in the next slot s_{k+1} in which the first group will transmit again (see also Fig. 4). Assume that the size of the first group created by splitting the original collision to be l_1 (and the second to be $i-l_1$). We consider four cases, elaborated in the following text.

Case 1: In this case it holds that $i > j$, implying that the size of the first group $l_1 < i$ and a collision of degree j can happen if the first group contends together with $j - l_1$ newly arrived users (denoted by the red arrow in the figure) in slot s_{k+1}. This mean number of collisions of degree j, denoted by $a_{i,j}^{(1)}$, is in this case given by

$$a_{i,j}^{(1)} = \sum_{l_1=0}^{j} \binom{i}{l_1} p^{l_1}(1-p)^{i-l_1} \ell_{j-l_1} = \sum_{l_1=0}^{j} b_{i,l_1} \ell_{j-l_1} \quad (1)$$

where $b_{i,l_1} = \binom{i}{l_1} p_1^l (1-p)^{i-l_1}$ is the probability that the size of the first group is l_1 and $\ell_{j-l_1} = \frac{\lambda^{j-l_1}}{(j-l_1)!} e^{-\lambda}$ is the probability that $j - l_1$ new users arrived.

Case 2: In the second case it is $i = j$, and a collision of degree j can occur if:

- the size of the first group is $l_1 < i$ and there were $i - l_1$ newly arrived users in the next slot,
- or $l_1 = i$ (i.e., all i users have chosen the first group) and there were $0 < k \leq K$ newly arrived users that were decoded after type-1 F-SIC was applied.

The expected number of collision of degree j is in this case

$$a_{i,j}^{(1)} = \sum_{l_1=0}^{i-1} b_{i,l_1} \ell_{i-l_1} + \sum_{k=0}^{K} b_{i,i} \ell_k \quad (2)$$

$$= \sum_{l_1=0}^{i} b_{i,l_1} \ell_{i-l_1} + \sum_{k=1}^{K} b_{i,i} \ell_k. \quad (3)$$

Case 3: In the third case $i < j \leq i + K$, and a collision of degree j can happen if $l_1 < i$ and there were $j - l_1$ newly arrived users in the next slot. Thus,

$$a_{i,j}^{(1)} = \sum_{l_1=0}^{i-1} b_{i,l_1} \ell_{j-l_1} = \sum_{l_1=0}^{i} b_{i,l_1} \ell_{j-l_1} - b_{i,i} \ell_{j-i} \tag{4}$$

where in (4) we used that fact that if $l_1 = i$ then $1 < j - i \leq K$, which is an outcome that falls into the second subcase of Case 2, hence the substraction.

Case 4: In the fourth case $i + K < j$ and the expected number of collisions of degree j is simply

$$a_{i,j}^{(1)} = \sum_{l_1=0}^{i} b_{i,l_1} \ell_{j-l_1} \tag{5}$$

Combining (1) - (5), we get

$$a_{i,j}^{(1)} = \sum_{l_1=0}^{\min(i,j)} b_{i,l_1} \ell_{j-l_1} + h_{i,j} \tag{6}$$

where

$$h_{i,j} = \begin{cases} 0, & i > j \text{ or } j > i + K \\ \sum_{k=1}^{K} b_{i,i} \ell_k, & i = j \\ -b_{i,i} \ell_{j-i}, & i < j \leq i + K \end{cases} \tag{7}$$

By defining

$$\mathbf{B} = \begin{bmatrix} b_{K+1,0} & \dots & b_{K+1,K+1} & 0 & 0 & \dots & 0 \\ b_{K+2,0} & \dots & b_{K+2,K+1} & b_{K+2,K+2} & 0 & \dots & 0 \\ \vdots & & \vdots & \vdots & & \ddots & \vdots \\ b_{d,0} & \dots & b_{d,K+1} & b_{d,K+2} & \dots & & b_{d,d} \end{bmatrix} \tag{8}$$

and

$$\mathbf{L} = \begin{bmatrix} \ell_{K+1} & \ell_{K+2} & \dots & \ell_{d-1} & \sum_{l=d}^{\infty} \ell_l \\ \vdots & \vdots & \dots & \vdots & \\ \ell_0 & \ell_1 & \dots & \ell_{d-K-2} & \sum_{l=d-K-1}^{\infty} \ell_l \\ 0 & \ell_0 & \dots & \ell_{d-K-3} & \sum_{l=d-K-2}^{\infty} \ell_l \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \ell_0 & \sum_{l=1}^{\infty} \ell_l \\ 0 & 0 & \dots & 0 & \sum_{l=0}^{\infty} \ell_l \end{bmatrix} \tag{9}$$

we can write

$$\mathbf{A}^{(1)} = \mathbf{BL} + \mathbf{H} \tag{10}$$

where $\mathbf{A}^{(1)} = \left[a_{i,j}^{(1)}\right]$ and $\mathbf{H} = [h_{i,j}]$ are square matrices of order $d - K$.

The second child of the parent collision is created when the second group, which is of size $i - l_1$, where $i - l_1 \geq K$,[6] when its turn comes, splits into the first group of size l_2 and the second group of size $i - l_1 - l_2$, and contends with newly arrived users in slot s'_{k+1}, as depicted in Fig. 4.[7] The expected number of collisions of degree j in slot s'_{k+1} is

$$a_{i,j}^{(2)} = \sum_{l_1=0}^{i-K-1} b_{i,l_1} \left[\sum_{l_2=0}^{\min(i-l_1,j)} b_{i-l_1,l_2} l_{j-l_2} + h_{i-l_1,j} \right] \tag{11}$$

$$= \sum_{l_1=0}^{i-K-1} b_{i,l_1} a_{i-l_1,j}^{(1)} \tag{12}$$

where in (11) the factor b_{i,l_1} represents the probability that second group created in s_k is of size $i - l_1$, and the factor in square brackets is derived using the same approach used for the characterization of the collision in slot s_{k+1}. By defining

$$\mathbf{R} = \begin{bmatrix} b_{K+1,0} & 0 & 0 & \dots & 0 \\ b_{K+2,1} & b_{K+2,0} & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & & \vdots \\ b_{d,d-K-1} & b_{d,d-K-2} & b_{d,d-K-3} & \dots & b_{d,0} \end{bmatrix} \tag{13}$$

we can write

$$\mathbf{A}^{(2)} = \left[a_{i,j}^{(2)}\right] = \mathbf{R}\mathbf{A}^{(1)} = \mathbf{R}(\mathbf{BL} + \mathbf{H}). \tag{14}$$

Continuing with this line of reasoning, it can be shown that, for the child collision in slot s''_{k+1} we have

$$\mathbf{A}^{(3)} = \left[a_{i,j}^{(3)}\right] = \mathbf{R}\mathbf{A}^{(2)} = \mathbf{R}^2(\mathbf{BL} + \mathbf{H}) \tag{15}$$

where $a_{i,j}^{(3)}$ is the number of ways that an initial collision of degree i will result in a third child collision of degree j. In general, we have

$$\mathbf{A}^{(r)} = \left[a_{i,j}^{(r)}\right] = \mathbf{R}\mathbf{A}^{(r-1)} = \mathbf{R}^{r-1}(\mathbf{BL} + \mathbf{H}) \tag{16}$$

where $a_{i,j}^{(r)}$ is the probability that an original collision of degree i will result in a collision of degree j in its r-th child.

Thus, the total expected number of collisions of degree j that are children of the parent collision of degree i, $K \leq i, j \leq d$, is

$$m_{i,j} = \sum_{r=1}^{\infty} a^{(r)} \tag{17}$$

and the matrix $\mathbf{M}$ is given by

$$\mathbf{M} = [m_{i,j}] = \sum_{r=1}^{\infty} \mathbf{A}^{(r)} = \sum_{r=0}^{\infty} \mathbf{R}^r \left(\mathbf{BL} + \mathbf{H}\right) \tag{18}$$

$$= (\mathbf{I} - \mathbf{R})^{-1}(\mathbf{BL} + \mathbf{H}) \tag{19}$$

where the convergence of the geometric series in (18) stems from the fact that (i) $\mathbf{R}$ is a lower triangular matrix, therefore, its elements on the main diagonal $b_{i,0} = (1-p)^i$, $i = K + 1, \dots, d$, are its eigenvalues, and (ii) its easy to verify that all the eigenvalues are lower than 1.

[6] If $i - l_1 \leq K$, then the users in this group will decoded after backward SIC is applied.

[7] Note that s'_{k+1} does not necessarily has to immediately follow s_{k+1}, but the collision in it has the same parent and belongs to the same generation as the one in s_{k+1}.

It is easy to show that $\mathbf{M}$ is a positive and irreducible matrix. Thus, its dominant eigenvalue is positive and equal to its spectral radius $\rho(\mathbf{M})$. Moreover,

$$\rho(\mathbf{M}) \leq \max_{K<j\leq d} \sum_{i=K+1}^{d} m_{i,j} \tag{20}$$

see [14, Theorem 8.1.22.], which is the result that we will make use of in Section IV.

IV. Upper bound on MST of Variant B

An exact analysis of variant B is essentially intractable, due to the fact that potentially all generation of the collisions in the tree may get entangled and the decoding of the users involved in them become interdependent. This can be seen by comparing the examples in Fig. 1 and Fig. 2, where the child collision in slot 5 is of a degree 5 in the former case and of degree 4 in the latter, due to user E being resolved in a collision that occurred in slot 2 (i.e., two generations before) and removed from the parent collision in slot 3 via application of forward SIC. As a consequence, the theory of MTP processes can not be applied.

Still, we can provide some insights on the spectral radius of the matrix $\widetilde{\mathbf{M}}$, the analogue of matrix $\mathbf{M}$ derived in Section **??**. Specifically, $\widetilde{\mathbf{M}}$ would be given by a sum analogous to the one in (18)

$$\widetilde{\mathbf{M}} = [\tilde{m}_{i,j}] = \sum_{r=1}^{\infty} \widetilde{\mathbf{A}}^{(r)} \tag{21}$$

where $\widetilde{\mathbf{A}}^{(r)}$ is the analogue of matrix $\mathbf{A}^{(r)}$ and whose entries are $\tilde{a}_{i,j}^{(r)}$, $K < i, j \leq d$.

$\widetilde{\mathbf{A}}^{(1)}$ concerns the first child collision (of the degree greater than K), which happens in the slot that immediately follows slot with the parent collision, see Fig. 2 and slots 3 and 4. Since this collision is undecodable, there is no chance for it to become decoded via type-2 F-SIC (there were no slots in between and no decodings possible), also implying that $\widetilde{\mathbf{A}}^{(1)} = \mathbf{A}^{(1)}$.

$\widetilde{\mathbf{A}}^{(2)}$ concerns the second child collision, which occurs in a slot that does not immediately follow the one of the parent collision, allowing for the type-2 F-SIC to take place. This is illustrated in Fig. 2, where the child collision slot 5 is affected by the forward SIC that removed user E from the parent collision in slot 3 (and thus also from the second group created in slot 3). In general, due to potential for type-2 B-SIC and type-2 F-SIC to happen in this case, the parent collision of degree i will become of degree $i-n$, $0 \leq n \leq i$, with some probabilities $\alpha_{i,n}$, where $\sum_{n=0}^{i} \alpha_{i,n} = 1$. If $i - n \leq K$, then the second group will be immediately decoded (this is the case for the second group {C,F} created in slot 2 in Fig. 2 and there will be no children that follow. The probability for this to happen is simply $\beta_i = \sum_{n=i-K}^{i} \alpha_{i,n} \leq 1$. Thus, we can write

$$\tilde{a}_{i,j}^{(2)} = \alpha_{i,0}\, a_{i,j}^{(2)} + \alpha_{i,1}\, a_{i+1,j}^{(2)} + \cdots + \alpha_{d,d-i}\, a_{d-i,j}^{(2)} \tag{22}$$

$$= \sum_{n=0}^{d-i} \alpha_{i+n,n}\, a_{i+n,j}^{(2)} \tag{23}$$

for $K < i \leq d$, and where $a_{i,j}^{(2)}$ are given by (12). The sum of the j-th column, $K < j \leq d$, of matrix $\widetilde{\mathbf{A}}^{(2)}$ is

$$\sum_{i=K+1}^{d} \tilde{a}_{i,j}^{(2)} = \sum_{i=K+1}^{d} \sum_{n=0}^{d-i} \alpha_{i+n,n}\, a_{i+n,j}^{(2)} \tag{24}$$

$$= \sum_{i=K+1}^{d} \sum_{n=0}^{i-K-1} \alpha_{i,n}\, a_{i,j}^{(2)} \tag{25}$$

$$= \sum_{i=K+1}^{d} (1-\beta_i)\, a_{i,j}^{(2)} \leq \sum_{i=K+1}^{d} a_{i,j}^{(2)} \tag{26}$$

i.e., this sum can not be larger than the sum of the corresponding column of $\mathbf{A}^{(2)}$, and in fact it is smaller if any of probabilities β_i, $K < i \leq d$, is non-zero[8]. Via analogous reasoning, the same conclusion can be established for the column sums of $\widetilde{\mathbf{A}}^{(r)}$ and $\mathbf{A}^{(r)}$, $r \geq 2$, i.e.,

$$\sum_{i=K+1}^{d} \tilde{a}_{i,j}^{(r)} \leq \sum_{i=K+1}^{d} a_{i,j}^{(r)}. \tag{27}$$

From (18), (21), and (27),it follows that for any j, $K < j \leq d$, we have

$$\sum_{i=K+1}^{d} \tilde{m}_{i,j} \leq \sum_{i=K+1}^{d} m_{i,j}. \tag{28}$$

Thus, we have that

$$\rho(\widetilde{\mathbf{M}}) \leq \max_{K<j\leq d} \sum_{i=K+1}^{d} \tilde{m}_{i,j} \leq \max_{K<j\leq d} \sum_{i=K+1}^{d} m_{i,j}. \tag{29}$$

In other words, the upper limit on spectral radius of $\widetilde{\mathbf{M}}$ can not be larger than the one of $\mathbf{M}$, see (20), and if any of the probabilities β_i, $K < i \leq d$ is non-zero, then it is strictly smaller.

Subcriticality and rate of extinction vs criticality

V. Evaluation

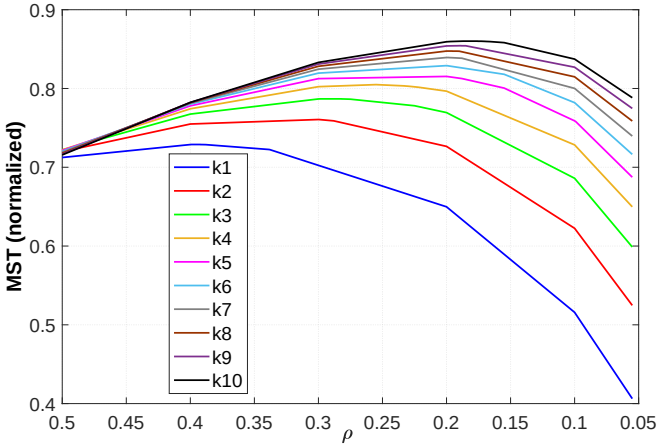


Fig. 5. Performance evaluation of K-MPR with SIC in the infinite memory BTA.

[8]This is reasonable to assume.

TABLE I
EVALUATION OF THE METRICS BASED ON MST FOR INFINITE BTA

k	ρ	MST	$E[T]$	$E[D]$	$E[W]_{\zeta=.8}$
1	0.3921	0.728853875	1.2512	1.7512	2.6521
2	0.2994	1.521146231	1.0417	1.5417	2.2751
3	0.2819	2.360256931	1.0075	1.5075	2.2134
4	0.2327	3.212958123	1.0007	1.5007	2.2013
5	0.1998	4.076342638	1.0001	1.5001	2.2001
6	0.1991	4.973157862	1.0000	1.5000	2.2000
7	0.1982	5.874779432	1.0000	1.5000	2.2000
8	0.1955	6.779634873	1.0000	1.5000	2.2000
9	0.1886	7.688294889	1.0000	1.5000	2.2000
10	0.1757	8.600664392	1.0000	1.5000	2.2000

VI. CONCLUSIONS

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